

Ease: An R Package for Efficient Population Genetics Simulations

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Summary

Ease is an R package for efficient and reproducible population genetics simulations, designed to facilitate the implementation of theoretical models involving few to moderate numbers of loci with arbitrary epistasis and dominance patterns. The package implements efficient simulation of such models by tracking genotype frequencies using deterministic transition matrices for mutation, reproduction, and selection. It incorporates genetic drift via multinomial sampling. Ease provides a flexible simulation environment well suited to theory-driven research in population genetics.

Statement of need

Population genetics plays a crucial role in our understanding of the adaptation of species. Its methods are used, for example, to trace back speciation times, to understand the genetic diversity of species and the consequences of selection. The field has always been largely driven by theory, which, thanks to simplified models of reality, has made it possible to understand and predict how evolution shapes biodiversity. Along with mathematical treatment of those models, simulations are essential, particularly for exploring analytically intractable models or for assessing their behaviour when their assumptions are relaxed. A wide range of simulation frameworks are available, e.g. coalescent simulators optimised for neutral genome-scale models (e.g. [msprime](#); [@baumdicker_2022](#)), or forward-in-time individual-based simulators designed for biologically realistic representations of selection, demography and drift (e.g. [SLiM](#); [@Haller_2023](#)). These tools are well suited to many applications, but they are not optimised for models in which the genetic architecture is deliberately simple while selection depends on complex epistasis and dominance relations between finite set of alleles in arbitrarily large populations. Many canonical models (see, for example, models described in [@crow_introduction_1970](#); [@burger_mathematical_2000](#); [@ewens_mathematical_2004](#)) fall into this category, which have not been the focus of efforts to produce an effective, flexible and easy-to-use simulator.

The R package Ease was developed to fill this gap. It is a lightweight and reproducible implementation of population genetics models that include only a few loci. Ease represents populations by genotype frequency vectors rather than by individual genomes, and updates these frequencies across generations under selection, mutation, recombination, migration, and genetic drift. This approach is computationally efficient when the number of loci and alleles is limited, and it closely matches the structure of many population genetic models that are formulated as recursions on genotype or haplotype frequencies.

39 Software design

40 Ease incorporate both nuclear and cytoplasmic (homoplasmic) loci. The loci of the first type
41 are diploid, with Mendelian transmission, while those of the second type are haploid with a
42 strict maternal transmission. Any number of alleles can be defined for each locus. Unlike
43 individual-based simulations, where each individual is explicitly defined (i.e. each with its own
44 combination of alleles), Ease tracks the frequencies of allele combinations themselves. Allele
45 combinations are defined as unique sets of alleles, and exist in two types: *haplotypes* and
46 *genotypes*. A *haplotype* is defined as the allele combination of a gamete, which is composed of
47 one *nuclear haplotype* and one *cytoplasmic haplotype*. Two *nuclear haplotypes* can be paired
48 together to form a *nuclear genotype* and combined with a *cytoplasmic haplotype* to form a
49 *genotype*. A *genotype* is therefore the allele combination of a zygote.

50 Once the user has defined how many loci (nuclear and cytoplasmic) and how many alleles per
51 locus they want to simulate, Ease defines automatically a set of transition matrices which are
52 used at each generation and constant throughout the simulation (Figure [Figure 1](#)). These
53 matrices allow, for example, the haplotype frequency vector of gametes to be calculated from
54 the genotype frequencies of the parents by a simple matrix multiplication that incorporates
55 recombination, random segregation of chromosomes, and mutation. Theoretically, no hard
56 limit is imposed onto the number of loci and the number of alleles per locus the user can
57 define, but the computational method Ease uses to track allele frequency changes imposes a
58 practical limit to them. This is because the size of the transition matrices becomes exceedingly
59 large when the number of haplotypes and the number of genotypes is large, and so when the
60 number of loci exceeds a handful. One advantage of this approach is that populations can
61 be of any size (up to millions or billions of individuals) without slowing down the simulation.
62 This is because individuals are not simulated explicitly. Genetic drift is instead simulated by
63 multinomial sampling, whose performance is only marginally affected by sample size.

64 The primary use of Ease is to track the trajectories of genotypes, haplotypes and allele
65 frequencies over generations. It does so through a canonical life cycle used in population
66 genetics, in which diploid zygotes are first produced by the fusion of one male and one female
67 haploid gametes resulting from independent meiosis (Figure [Figure 1](#)). These zygotes then
68 develop into adults, which then produce the gametes that will form the zygotes of the next
69 generation before dying (generations do not overlap). Along this life cycle, selection can
70 happen at three time points. Adult genotypes can be associated with absolute fecundity values
71 for female and male gamete production. Gametes, depending on their haplotype and their sex,
72 can be associated with survival probabilities. Finally, zygote genotypes can be associated with
73 different survival probabilities, which occur before they become adults.

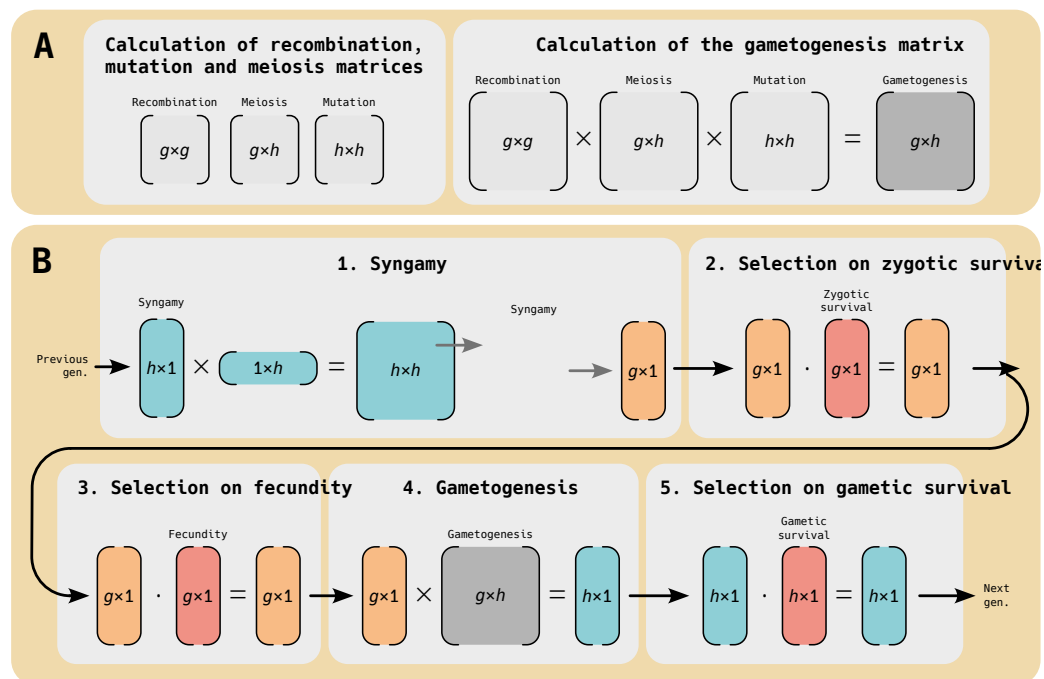


Figure 1: Summaries of the various steps implemented in Ease to run simulations with a number G of genotypes and H of haplotypes. All the transition matrices (reproduction, selection, migration, ...) are available for consultation using the print method, and can be defined in a variety of ways, depending on user's requirements.

Ease assumes two types of gamete associated with two mating types, which are called male and female gametes for simplicity. For the sexual identity of adult individuals, there are two possible scenarios. In the default scenario, all individuals are hermaphrodites and therefore capable of producing both female and male gametes. A self-fertilisation rate can be define, as the probability for a zygote to be produced through the fusion of gametes produced by the same individual. In an alternative scenario, zygotes are assigned to develop into adults producing only male gametes or only female gametes, with a 50% chance for each. It is possible to simulate this alternative scenario from the default one by defining explicitly a sex-determining locus, which can be useful when simulating sex-linked loci or variable sex ratios. This can also help to simulate more complex sexual systems, such as gynodioecy or androdioecy, among others.

Finally, the package supports the simulation of multiple populations with any migration pattern between them, e.g. island models, stepping stone models, or models with different population sizes.

Example of use

Model implementation is standardised and involves defining the genome, allelic effects, population(s) and their interconnection within a metapopulation, before finally simulating and extracting the results of the simulation(s) (Figure 2). A model can be simulated multiple times (without parallelisation), their results are then combined in the result objects.

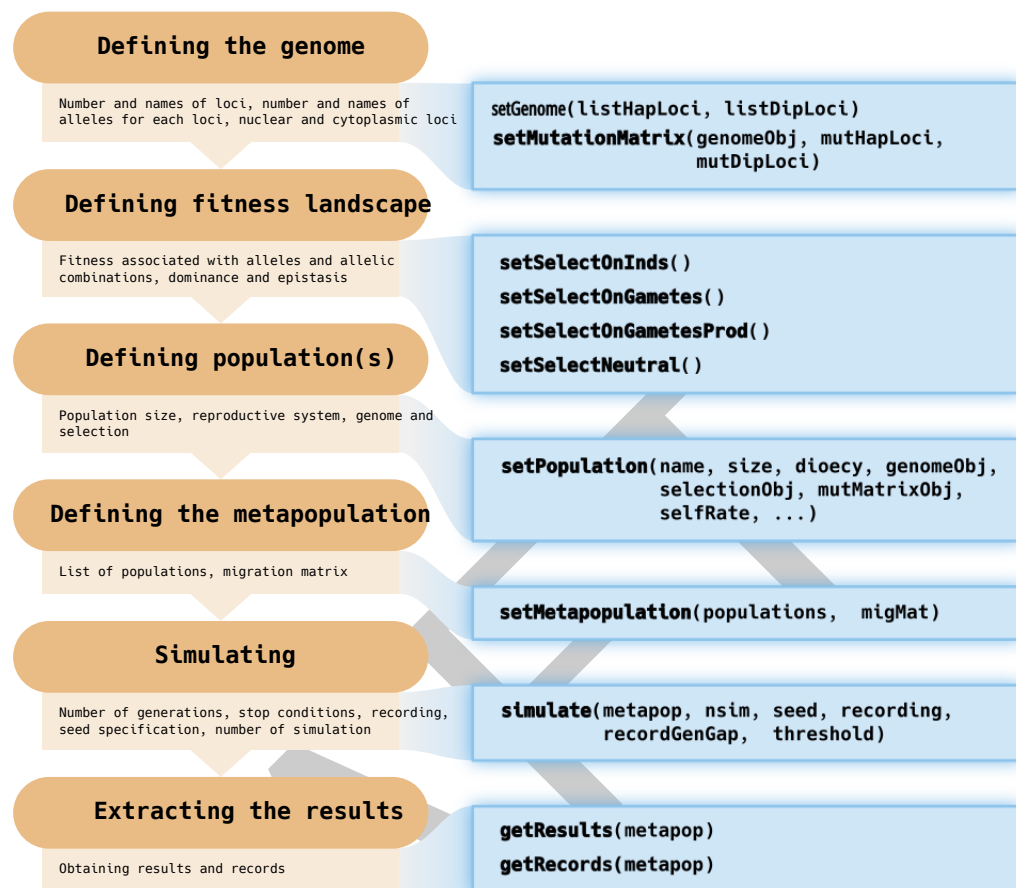


Figure 2: Steps for implementing a model with Ease and associated functions, from defining the genome and the selective effect of alleles to extracting the results of the model simulation.

93 In the example below, each individual is represented by a genotype at two diallelic loci. One of
94 these is nuclear with wildtype allele "A" and alternative allele "a", and the other is cytoplasmic
95 with wildtype allele "B" and alternative allele "b". By default, when defining a population,
96 the first allele named (here the wildtypes "A" and "B", see below) is the one that is fixed
97 in the population at the start of the simulation. {r} library(Ease) # Definition of a
98 genome with 2 locus, 2 alleles each: DL <- list(dL = c("A", "a")) HL <- list(hL
99 = c("B", "b"))

100 Next, a list of possible mutation events from one allele to another is defined, each
101 associated with a mutation rate, defined as the proportion of copies that will be
102 mutated at each meiosis event. {r} mutations <- list(mutation(from =
103 "A", to = "a", rate = 1e-3), mutation(from = "B", to = "b", rate = 1e-3)
104) genomeObj <- Ease::setGenome(listHapLoci = HL, listDipLoci = DL) mutMatrixObj
105 <- Ease::setMutationMatrix(genomeObj, mutations = mutations) The population is then
106 defined, specifying its name (here ExamplePop), size (here 1,000 individuals), sexual system
107 (here dioecy, i.e. 500 females and 500 males), genetic system and mutation pattern. In addition
108 to this, it is necessary to define the type of selection. Here, for simplicity's sake, the alleles are
109 defined as neutral. {r} # Definition of a population of 1000 individuals, males and
110 females pop <- Ease::setPopulation(name = "ExamplePop", size = 1000,
111 dioecy = TRUE, genomeObj = genomeObj, mutMatrixObj = mutMatrixObj,
112 selectionObj = Ease::setSelectNeutral(genomeObj)) print(pop) #> ----
113 = Population OBJECT ---- #> in details #> Population
114 'ExamplePop' of 1000 dioecious individuals #> There is no demography. #> The

```

115 initial genotypes frequency are:  #> A/A||B A/a||B a/a||B A/A||b A/a||b a/a||b
116 #>      1      0      0      0      0      0 #> Selection:  #> No selection
117 defined.  #> ----- Once the population has been
118 defined, a metapopulation must be defined (here with only a single population). Below,
119 its evolution is simulated over 2,000 generations with the genotype, haplotype and allele
120 frequencies of each generation being recorded (by default when setting recording =
121 TRUE). {r} metapop <- Ease::setMetapopulation(populations = list(pop)) metapop
122 <- simulate(metapop, threshold = 2000, seed = 123, recording = TRUE) rec =
123 Ease::getRecords(metapop) print(which(rec$s1$ExamplePop$a == 1)[1]) #> [1] 1072
124 print(which(rec$s1$ExamplePop$b == 1)[1]) #> [1] 1481 The recorded frequencies of
125 alleles a and b can then be extracted using the function getRecords. Here, having defined a
126 (fairly high) directional mutation rate towards these alleles, each fixed after 1072 and 1481
127 generations, respectively.

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128 For more details on how the package works and a more detailed view of the options, see the
129 package vignette and documentation.

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132 modelling approach, and Vitor Sudbrack for his valuable feedback.

133 References